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Chemical and Electrochemical O_2 Reduction on Earth-Abundant M-N-C Catalysts and Implications for Mediated Electrolysis

Jason S. Bates[1], **Sourav Biswas**[1,Δ], **Sung-Eun Suh**[1,‡], **Mathew R. Johnson**[1], **Biswajit Mondal**[1], **Thatcher W. Root**[2,*], **Shannon S. Stahl**[1,*]

[1]Department of Chemistry, University of Wisconsin–Madison, 1101 University Avenue, Madison, WI 53706, USA

[2]Department of Chemical and Biological Engineering, University of Wisconsin–Madison, 1415 Engineering Drive, Madison, WI 53706, USA

Abstract

M-N-C catalysts, incorporating non-precious metal ions (e.g., M = Fe or Co) within a nitrogen-doped carbon support, are the focus of broad interest for electrochemical O_2 reduction and aerobic oxidation reactions. The present study explores the mechanistic relationship between the O_2 reduction mechanism under electrochemical and chemical conditions. Chemical O_2 reduction is investigated via aerobic oxidation of a hydroquinone, in which the O–H bonds supply the protons and electrons needed for O_2 reduction to water. Mechanistic studies have been conducted to elucidate whether the M-N-C catalyst couples two independent half-reactions (IHR), similar to electrode-mediated processes, or mediates a direct inner-sphere reaction (ISR) between O_2 and the organic molecule. Kinetic data support the latter ISR pathway. This conclusion is reinforced by rate/potential correlations that reveal significantly different Tafel slopes, implicating different mechanisms for chemical and electrochemical O_2 reduction.

Graphical Abstract

[*]**Corresponding Authors**: twroot@wisc.edu; stahl@chem.wisc.edu.
[Δ]S.B., present address: Department of Chemistry, Buffalo State, State University of New York, 1300 Elmwood Ave., Buffalo, NY, 14222, USA
[‡]S.-E.S., present address: Department of Chemistry, Ajou University, Suwon 16499, Republic of Korea.

Supporting Information. Experimental procedures, characterization data, kinetic data, adsorption data, rate expression derivations, and supplementary discussions. This material is available free of charge via the internet at http://pubs.acs.org.

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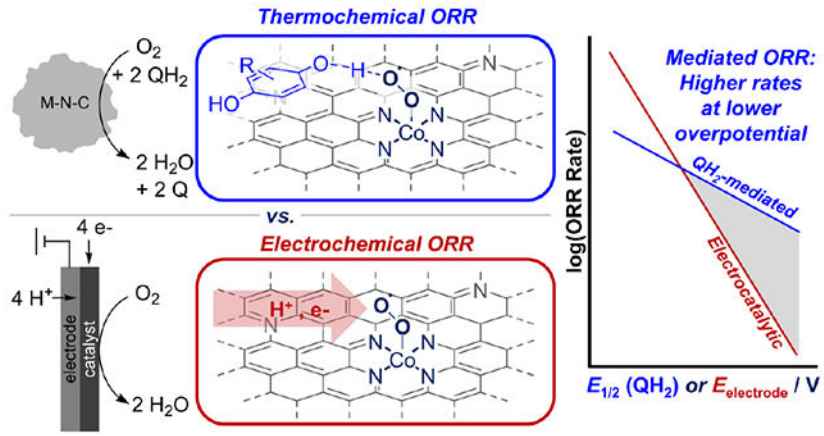

INTRODUCTION

Heterogeneous catalysts play a crucial role in reactions with molecular oxygen, ranging from the oxygen reduction reaction (ORR) in fuel cells[1–3] to the selective oxidation of organic molecules[4–8] and direct synthesis of hydrogen peroxide.[9,10] In fuel cells, electrocatalytic O_2 reduction and H_2 oxidation take place at separate electrodes, with electrons and protons transferred through an electrical circuit and ion-exchange membrane, respectively (Figure 1a–i). Thermochemical reactions between H_2 or an organic molecule ($SubH_2$) and O_2 are distinct from their electrochemical counterparts because the substrates are colocalized and react at the same catalyst (Figure 1a–ii). These physical differences, however, conceal the prospect for mechanistic similarities between electrochemical and thermochemical reactions with O_2.[11–16] For example, recent studies by Surendranath[15] and Flaherty and Rodríguez-López[16] have investigated mechanisms of reactions of O_2 with formic acid and ethanol[15] and with H_2[16] using noble metal catalysts. The data indicate that the catalysts mediate independent ORR and substrate oxidation redox half-reactions, formally reflecting a "short-circuited" electrochemical cell (Figure 1b–i).[17] Here, we report a mechanistic study of aerobic oxidation of a hydroquinone (**HQ1**[18]) using non-precious metal "M-N-C" catalysts (eq 1). These materials feature Fe, Co, or other first-row transition-metal ions integrated within a nitrogen-doped graphitic carbon support, and they have emerged as compelling alternatives to noble metal catalysts for electrochemical ORR.[19–22] The data presented below provide evidence for direct inner-sphere reaction (ISR) between O_2 and the organic molecule (Figure 1b–ii), [23, 24] contrasting the independent half-reaction (IHR) mechanism observed with noble-metal catalysts.

(1)

Mediated ORR systems represent an intersection between electrochemical and thermochemical reactions. [25] **HQ1** was recently paired with a heterogeneous Co-Phen-C [26] aerobic oxidation catalyst to achieve mediated electrochemical ORR (Figure 1c), and it supported current densities much higher than previous mediated ORR systems.[27,28] The present study of M-N-C-catalyzed O_2 reduction by **HQ1** provides unique insights into the relationship between mediated and direct electrocatalytic ORR.

RESULTS AND DISCUSSION

Kinetic studies of **HQ1** oxidation to **Q1** were initiated under the condition of eq 1, with kinetically limited initial rates quantified under differential conversion regimes in well-mixed batch reactors (conversion < 20%, rates normalized per gram of catalyst; see Section 4.2 of the Supporting Information for details, including evidence that rates are not limited by mass transport). Co-Phen-C was selected for this kinetic study based on its prior success in our **HQ1**-mediated ORR system.[27] M-N-C catalysts made with 1,10-phenanthroline are reported to generate atomically dispersed MN_x moieties, often in combination with metal nanoparticles. [29–31] The single-atom MN_x sites have been implicated as the active sites for both electrocatalytic ORR[19–22] and thermal aerobic oxidations.[30,32] Our own observations are consistent with this conclusion. For example, several Co-Phen-C catalyst samples were prepared that contain different total cobalt loading (1.4–3.0 wt% Co), and powder X-ray diffraction data reveal the presence of Co nanoparticles (see section 3 of the Supporting Information). Acid washing (H_2SO_4), which selectively removes nanoparticles,[30] lowers the total Co loading; however, the untreated and acid-washed catalyst exhibit very similar rates per gram of catalyst in the oxidation of **HQ1**. These control experiments, elaborated in section 4.3.2 of the Supporting Information, support catalysis by CoN_x moieties rather than by nanoparticles. We note that the kinetic data and mechanistic conclusions presented herein do not rely on a particular active-site identity, but these considerations account for our schematic representation of active sites as atomically dispersed MN_x sites.

The partial pressure of O_2 and concentrations of **HQ1**, H^+, and **Q1** were varied relative to a set of reference conditions (1.1 bar O_2, 0.5 M H_2SO_4, 50 mM **HQ1**, 1 mM **Q1**) to determine the kinetic reaction orders (Figure 2). The rate of **HQ1** oxidation is first-order in O_2, but zero-order in [**HQ1**] (Figure 2a and 2b), reflecting the simplified rate expression in eq 2, where $[O_2]$ is the concentration of O_2 in solution (M), and k_1 is the apparent

$$r_{ox} = k_1[O_2] \quad (2)$$

first-order rate constant in O_2 (mol (M O_2)$^{-1}$ g^{-1} s^{-1}). Values of k_1 are zero-order in H^+ concentration between 0.05–0.5 M. The rates are slower when the reaction is conducted in D_2O, but they remain independent of the D^+ concentration (Figure 2c, for consideration of O_2 solubility effects, see Figure S13 in the Supporting Information). The difference between the rates in H_2O and D_2O corresponds to a primary H/D kinetic isotope effect (KIE) of 2.4 ± 0.4 (Figure 2c). Finally, product inhibition is evident from measuring the rate at different [**Q1**]/[**HQ1**] ratios (Figure 2d), with slower rates becoming apparent at [**Q1**]/[**HQ1**] > 1.

These kinetic data offer insights into the turnover-limiting step and catalyst resting state and provide preliminary support for inner-sphere reactivity over independent half-reactions (ISR and IHR in Figure 1b), according to the following analysis. The first-order O_2 dependence (Figure 2a) supports a turnover-limiting reaction with O_2 and indicates that O_2 is not chemisorbed at MN_x sites in the resting state. Both IHR and ISR pathways could be consistent with these features; however, the O_2 dependence requires IHR pathways with turnover-limiting ORR steps and quasi-equilibrated **HQ1**/**Q1** redox steps. The H/D KIE value of 2.4 is similar to a KIE value that has been measured for ORR with M-N-C electrocatalysts[33] and, thus, is consistent with an IHR model where ORR is rate-controlling. The lack of [H^+] dependence could be consistent with the IHR model, but only for a narrow set of kinetic criteria for ORR and **HQ1**/**Q1** redox steps (see eq S46 and the rate law derivation in Section 8.2 of the Supporting Information). The zero-order [**HQ1**] dependence and onset of product inhibition at high [**Q1**] is inconsistent with the IHR mechanism, as IHRs should exhibit a linear log-log dependence on [**Q1**]/[**HQ1**] (see Section 8.2 and Figure S29 in the Supporting Information).

The ISR mechanism does not face similar complications. The kinetic dependence on [O_2] and [H^+] and the H/D KIE in this mechanism is rationalized by endergonic pre-equilibrium chemisorption of O_2, followed by turnover-limiting hydrogen-atom transfer (HAT) from a physisorbed **HQ1** to the bound O_2 species. The product inhibition evident in Figure 2d is readily rationalized by a competition between **Q1** and **HQ1** for site occupancy on the M-N-C surface. Incorporation of these steps into a catalytic cycle for the ISR mechanism (Scheme 1) provides the basis for derivation of catalytic rate expressions, eq 3 (low [**Q1**]), and eq 4 (competitive **HQ1**/**Q1** binding), where K_H and K_Q are the adsorption equilibrium constants for **HQ1** and **Q1**, respectively, on the Co-Phen-C surface (see Section 8, Supporting Information for full rate law derivations).

$$r_{ox} = \frac{k_1 k_H [\boldsymbol{HQ1}][O_2]}{1 + K_H[\boldsymbol{HQ1}]} \tag{3}$$

$$r_{ox} = \frac{k_1[O_2]}{1 + K_Q K_H^{-1}[\boldsymbol{Q1}][\boldsymbol{HQ1}]^{-1}} \tag{4}$$

HQ1 adsorption on Co-Phen-C is quantified in two ways: by fitting the kinetic data at low [**Q1**] and independent measurement of **HQ1** adsorption isotherms in the absence of O_2. The K_H values determined from these methods, 4900 ± 1900 M^{-1} and 3400 ± 960 M^{-1}, respectively are within experimental error of each other (Figure 3; see Section 6 of the Supporting Information for details). The inhibition by **Q1** is reasonably described by eq 4, and fitting of the data in Figure 2d results in a $K_Q K_H^{-1}$ value of 0.16, indicating that **Q1** and **HQ1** adsorb with similar strength. This conclusion is validated by independent adsorption isotherm measurement (K_Q = 11,000 ± 1700 M^{-1}, Section 6, Supporting Information). Collectively, the kinetic and adsorption data are well aligned with the ISR mechanism and suggest that Co-N-C surfaces are saturated with **HQ1** at early stages of the reaction, and with mixed **HQ1**/**Q1** binding as **Q1** accumulates.

According to ISR mechanisms, O_2 and the organic molecules co-adsorb on the Co-N-C catalyst surface.[34] The aromatic **HQ1** structure should enable physisorption without blocking CoN_x O_2 binding sites, and the favorable **HQ1** binding will ensure that **HQ1** will be available to participate directly in O_2 reduction (Scheme 2a). The kinetic data presented above are consistent with a mechanism that closely resembles hydroquinone-mediated O_2 reduction with a molecular Co-salophen complex (Scheme 2b).[23,35] Similar HAT-mediated O_2 reduction mechanisms have been observed with other molecular catalysts (Fe-porphyrins with pendant quinol groups [36] and a Mn-salophen complex with co-catalytic hydroquinone [37]), in addition to enzymatic reactions with O_2.[38]

Gas uptake data reveal an **HQ1**:O_2 stoichiometry of 2:1, consistent with four-electron reduction of O_2 to H_2O without accumulations of H_2O_2. H_2O_2 may form as an intermediate from the two-electron oxidation of **HQ1**. Control experiments suggest that either Co-Phen-C-catalyzed disproportionation of H_2O_2 or oxidation of **HQ1** with H_2O_2 as the oxidant could account for the net 2:1 **HQ1**:O_2 stoichiometry of the reaction (see Section 7 of the Supporting Information for full data and discussion), but neither influences the overall ORR kinetics.

Despite good kinetic evidence favoring the ISR over the IHR pathway, we sought further data to distinguish between the two mechanisms. If aerobic oxidation of **HQ1** followed the "electrochemical" IHR pathway, turnover-limiting ORR steps should exhibit Tafel behavior equivalent to electrocatalytic ORR, with catalyst potential set by the hydroquinone mediator (see Section 8.2, Supporting Information). To test this possibility, the nominal turnover frequency[39–41] (TOF, per total metal) of hydroquinone oxidation by O_2 (i.e., ORR) was quantified for three representative Fe- and Co-N-C catalysts using a suite of hydroquinones with redox potentials from 0.518–0.744 V (**HQ1**–**HQ6**, Figure 4b, full rate data in Section 4.4, Supporting Information). A complementary set of kinetic data were obtained by measuring ORR rates as a function of electrode potential via rotating-disk voltammetry (Figure 4a). We note that while nominal TOF values (i.e., rates per total metal) are sensitive to the fraction of MN_x sites in a given M-N-C catalyst, the resulting Tafel-like slopes are not sensitive to the presence of inactive nanoparticle phases (see Figure S7, Supporting Information). The electrocatalytic ORR rates exhibit Tafel slopes of ~80 mV dec^{-1}, consistent with values reported for M-N-C catalysts in the ORR literature[42] (light gray dashed lines, Figure 4b; alternative axes for Tafel analysis are found in Figure S21—the axes of Tafel plots are inverted relative to those of LFER plots used to analyze thermochemical kinetics). The TOF values measured for reduction of O_2 with hydroquinones depend exponentially on the HQ redox potentials and exhibit a linear free energy relationship (LFER) with Tafel-like slopes of 200–240 mV dec^{-1} for the three different M-N-C catalysts (solid black lines, Figure 4b). These observations provide additional evidence against the participation of an IHR mechanism in the aerobic oxidation of HQ species with M-N-C catalysts. First, the kinetic data predict that the reduction of O_2 by HQ species should depend on the HQ redox potential, with a Tafel slope identical to that of electrocatalytic ORR (full derivation in Section 8.2, Supporting Information). In addition, qualitative differences are evident in relative catalyst performance for the two different ORR approaches: the Fe-N-C catalyst is most effective for electrocatalytic ORR, while the

Co-N-C catalysts show higher rates than Fe in the aerobic oxidation reaction. This difference merits further investigation, but it may be rationalized by different kinetically relevant steps and resting states of electrocatalytic and H_2Q-mediated ORR, as manifested in their different LFER/Tafel slopes.

For an ISR mechanism (Figure 1b–ii), the TOF/HQ redox potential LFER in Figure 4b may be interpreted according to the Bell–Evans–Polanyi (BEP) principle, [43,44] whereby the transition-state free energies for hydroquinone O–H bond cleavage correlate with the thermodynamic HQ/Q redox potentials. The similar slopes observed for M-N-C catalysts derived from different metals (Co, Fe) and nitrogen precursors (polyaniline, Phen) suggest that these catalysts operate by similar mechanisms. On the other hand, rate/redox potential scaling relationships for the electrocatalytic and thermocatalytic ORR processes have very different slopes (Figure 4a). This feature has important implications for mediated ORR because the sensitivity of the ISR mechanism to hydroquinone redox potential is much less than the sensitivity of electrocatalytic ORR to the electrode potential. This outcome implies that stable, high-potential mediators could provide a means to support ORR rates exceeding that of conventional electrocatalytic ORR at low overpotential. The pursuit of mediators capable of achieving this goal merits significant attention.

CONCLUSIONS

In summary, this work shows that hydroquinone-mediated O_2 reduction on M-N-C catalysts occurs by a mechanism that involves direct inner-sphere reaction between adsorbed hydroquinone and O_2 molecules. This ISR mechanism is distinct from the coupling of independent half-reactions that has been demonstrated recently for aerobic oxidation of various substrates on noble metal catalysts. Unique rate/redox potential correlations associated with hydroquinone-mediated O_2 reduction introduce a mechanistic strategy to circumvent constraints imposed by Tafel relationships for direct electrocatalysis. The atomistic insights gained from this study also contribute to complementary efforts to develop improved non-precious metal catalysts for selective aerobic oxidation of organic molecules.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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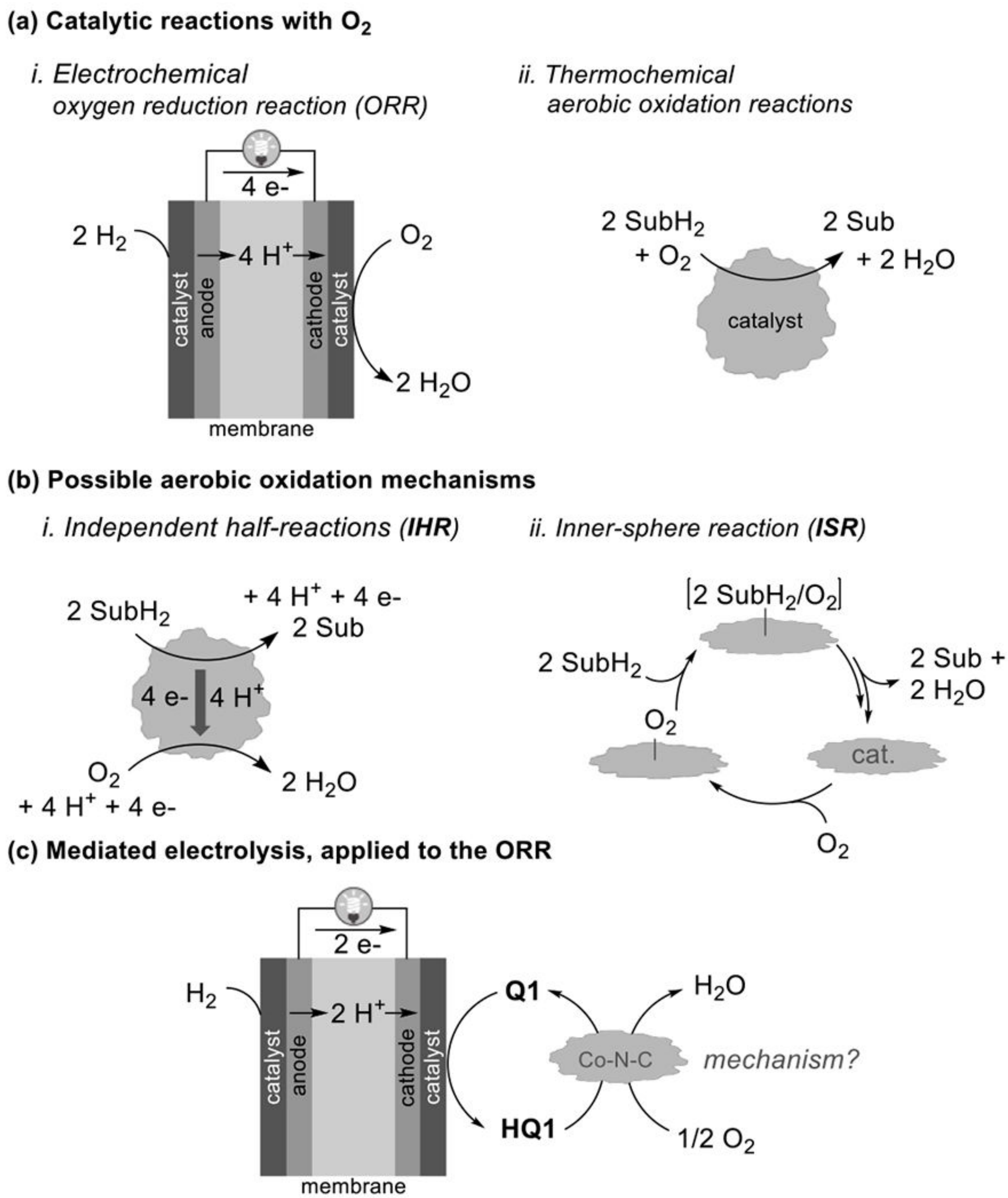


Figure 1.

Relationship between electrochemical and thermochemical reactions with O_2. (a) Relevance of O_2 reactivity to fuel cells and aerobic oxidation, (b) two potential reaction pathways for thermochemical reactivity of O_2 in aerobic oxidations, and (c) the intersection of electrochemical and thermochemical O_2 reactivity in mediated O_2 reduction.

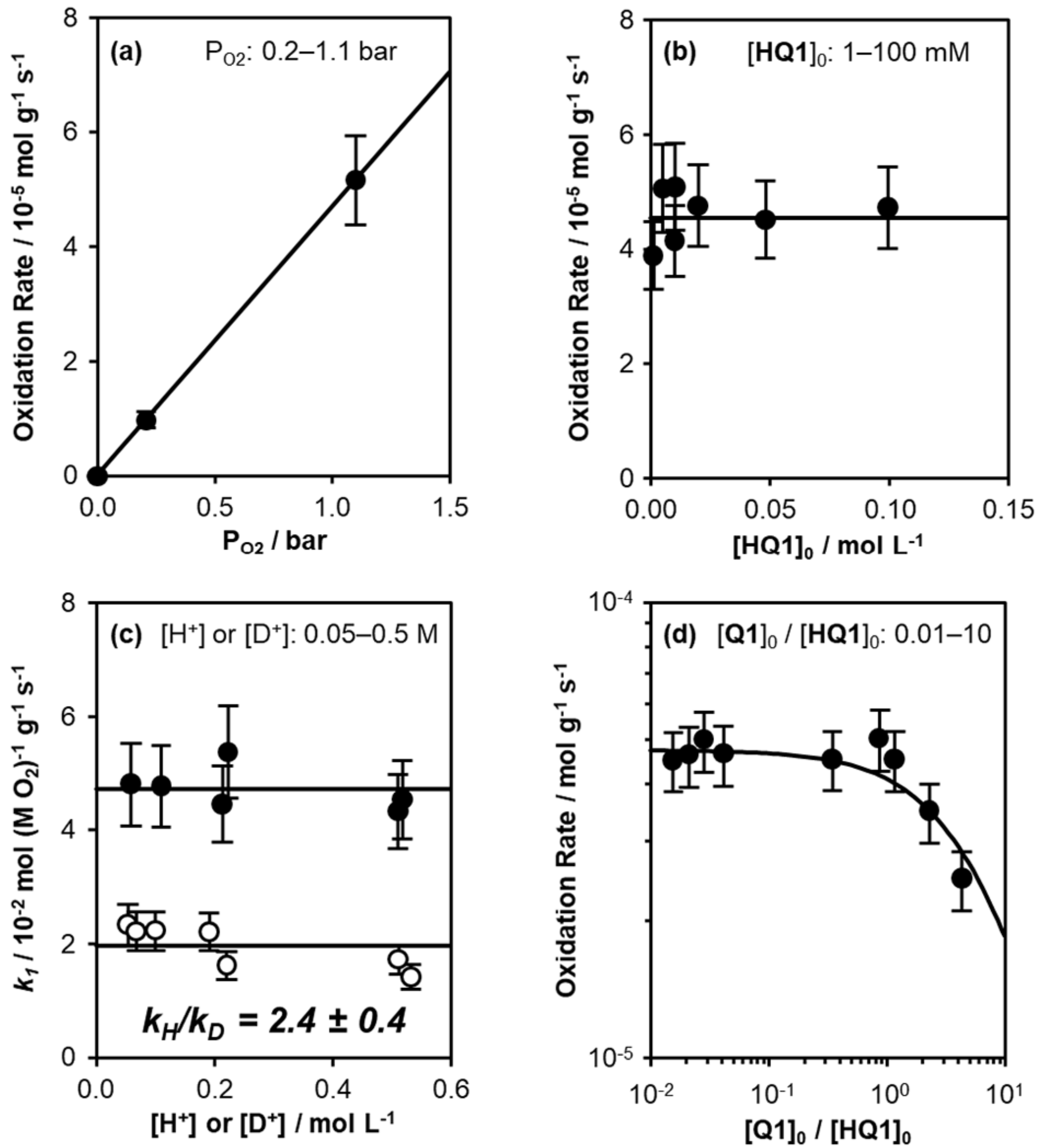


Figure 2.

Kinetics of **HQ1** oxidation on Co-Phen-C (1.8 wt% Co) at 30 °C. Standard conditions are 1.1 bar O_2, 0.5 M H_2SO_4, [**HQ1**]$_0$ = 50 mM, [**Q1**]$_0$ = 1 mM, except where noted for each panel as follows. (a) Oxidation rate as a function of O_2 pressure (0.2–1.1 bar). Solid line represents the best-fit linear regression to the data, constrained through the origin. (b) Oxidation rate as a function of **HQ1** concentration (1–100 mM; [**Q1**]$_0$/[**HQ1**]$_0$ < 0.1). Solid line represents best-fit regression to eq 3. (c) First-order rate constant (given by eq 2) as a function of [H^+] (0.05–0.5 M, filled symbols), or [D^+] (0.05–0.5 M, open symbols) in solutions prepared using D_2O. Solid lines represent the average values of each data set. (d) Oxidation rate as a function of [**Q1**]/[**HQ1**] ratio, where $[\mathbf{Q1}]_0$ is varied from 1–50 mM at constant $[\mathbf{HQ1}]_0$ = 50 mM, then $[\mathbf{HQ1}]_0$ is varied from 10–40 mM at constant $[\mathbf{Q1}]_0$ = 50 mM. Solid line represents best-fit regression to eq 4.

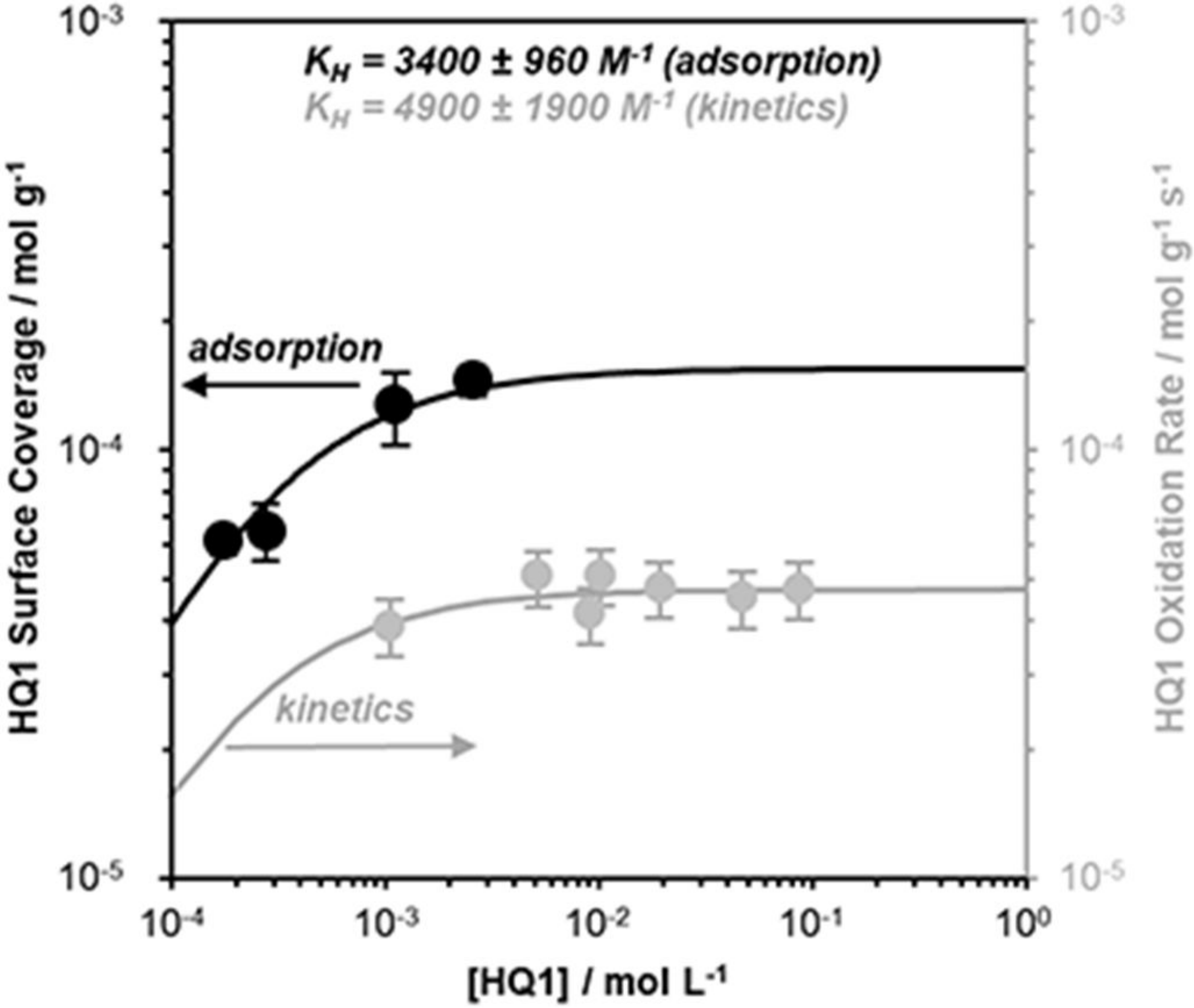
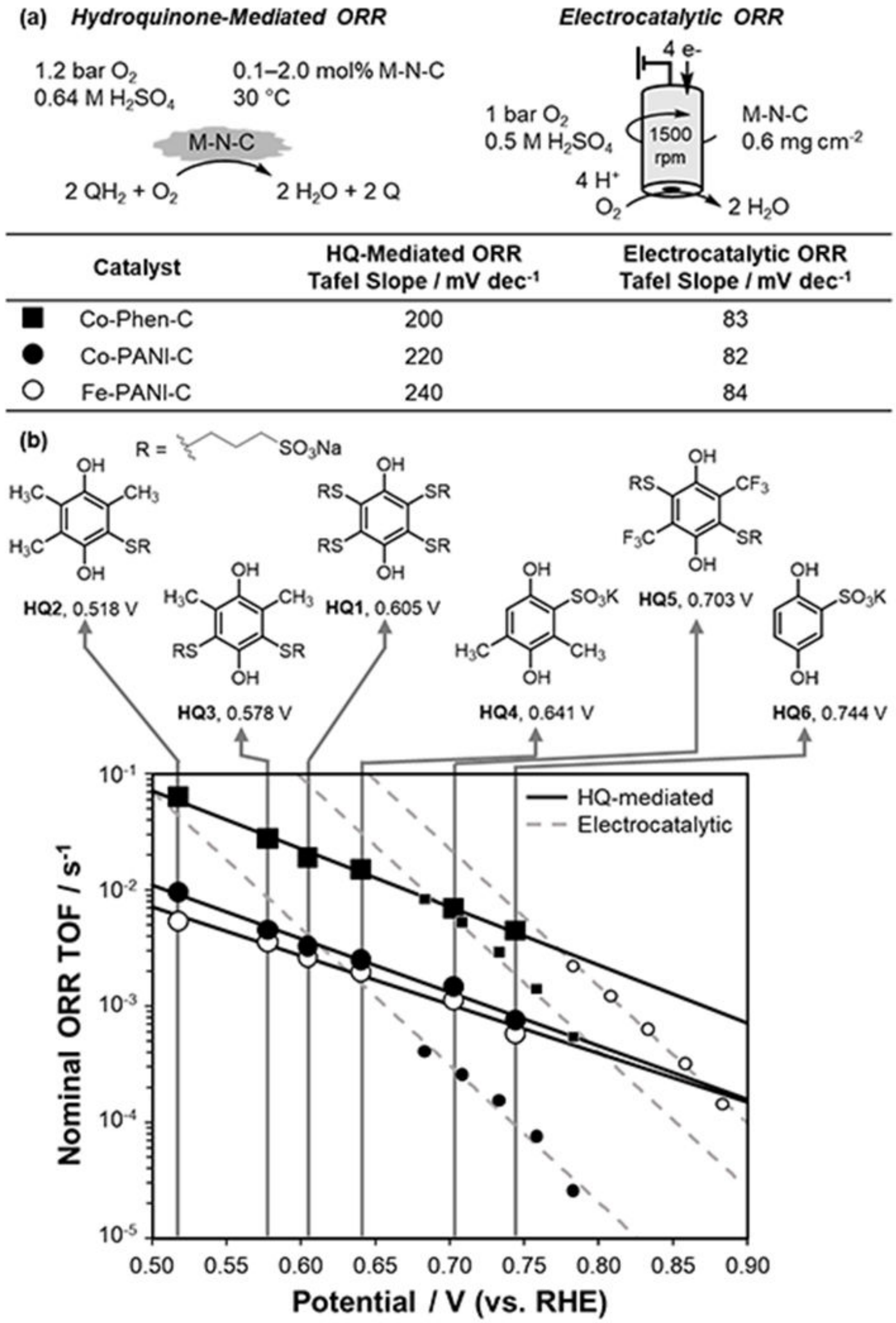


Figure 3.

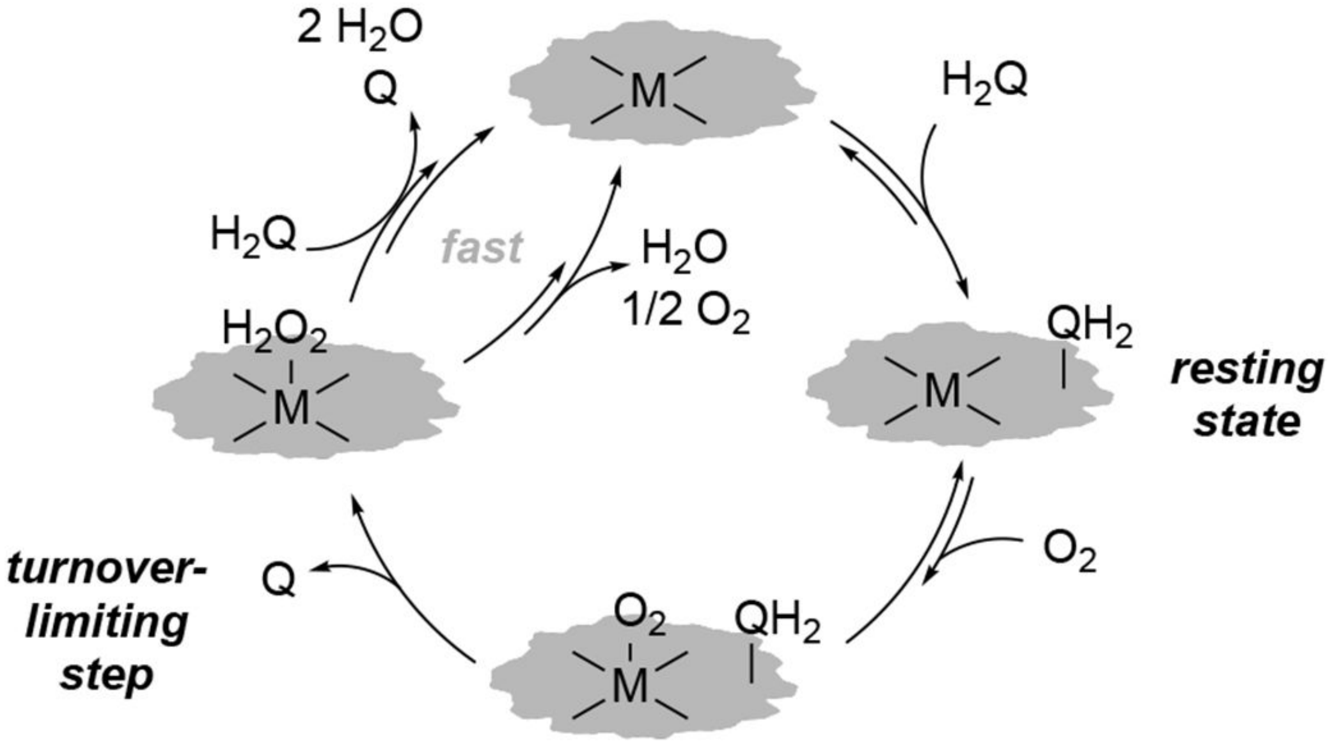
Comparison of the **HQ1** adsorption isotherm measured under N_2 (298 K) on Co-Phen-C (1.8 wt% Co) (black data points) with the kinetics of **HQ1** oxidation (gray data points, same data as Figure 2b). Solid lines reflect fits to the functional forms of eq S12 and eq 3, respectively.



Catalyst	HQ-Mediated ORR Tafel Slope / mV dec^{-1}	Electrocatalytic ORR Tafel Slope / mV dec^{-1}
■ Co-Phen-C	200	83
● Co-PANI-C	220	82
○ Fe-PANI-C	240	84

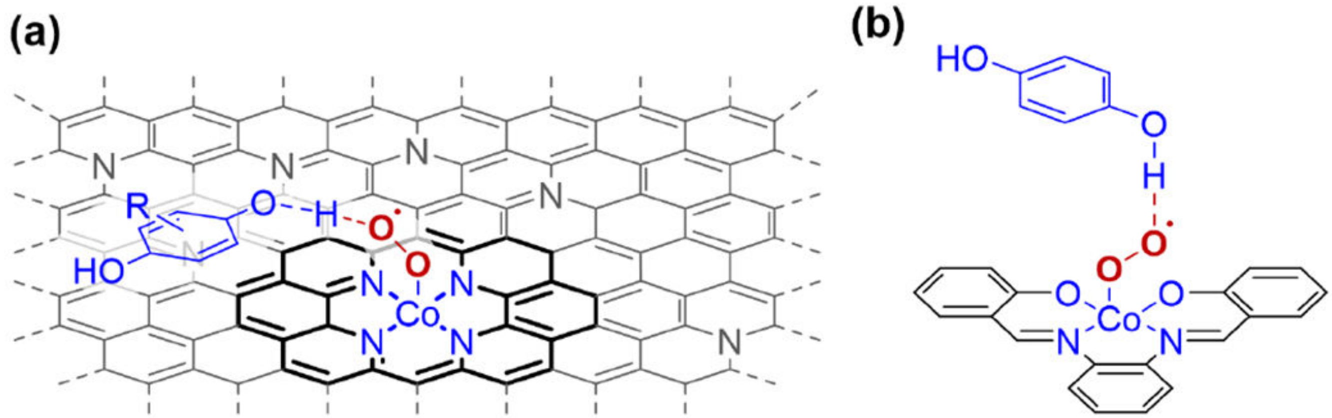
Figure 4.

Rate/potential relationships for HQ-mediated and direct electrocatalytic ORR. (a) Tafel slopes associated with the two different modes of M-N-C-catalyzed O_2-reduction. (b) Half-wave redox potentials of the different hydroquinones and plot of turnover frequencies (per total M) as a function of HQ redox potential (solid black line) and applied potential (dashed gray line) for Fe-PANI-C (○), Co-PANI-C (●) and Co-Phen-C (■) catalysts. Conditions: 1 mL of 0.05 M hydroquinone derivatives in 0.64 M H_2SO_4, catalyst (2.0 mol % Fe-PANI-C, 1.0 mol% Co-PANI-C and 0.1 mol% Co-Phen-C), 30 °C, 1.2 bar O_2. Kinetically limited steady-state electrocatalytic ORR TOF values measured by RDE (ambient T, 1500 rpm, O_2-saturated 0.5 M H_2SO_4, 0.6 mg cm^{-2}, n_e quantified by Koutecky–Levich analysis).



Scheme 1.

Proposed mechanism of inner-sphere O_2 reduction by HQ1.



Scheme 2.

Proposed HAT transition state for HQ1 oxidation on Co-Phen-C (a) and similar transition state proposed for hydroquinone oxidation by Co(salophen) (b).[23]